# The Power of "We" in Science Funding and Publication

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**Linguistic communication supports all stages of scientific innovation. However, we have limited understanding of rhetorical strategies for facilitating the recognition of new ideas in science. Drawing on prior literature on the importance of personal pronouns in inter-personal communication, we examine the unique role of first-person plural pronouns ("We" words, including "we", "us", "our") in science funding and publishing. Our study leverages the full sample of 13K grant proposals submitted to the Novo Nordisk Foundation and 47K research papers submitted to BMJO and ICLR between 2011 and 2024. We find a strong and universal positive association between "we" words and success rates even after accounting for important confounding factors related to content, novelty, and author impact. Specifically, a one percentage point increase in the percentage of "We" words used in the full text is associated with a 36-51% increase in the odds of success for grants and papers. Furthermore, we find that the effect of "We" is applicable to both editors and reviewers of BMJO papers that used the single-blind review and is predictive of more positive and engaging review feedback for ICLR papers that employed the double-blind review. Crucially, a large-scale LLM-assisted computational experiment that replaces "We" words with demonstrative pronouns provides robust causal evidence of "We" words on paper acceptance. Lastly, our causal mediation analysis shows that "We" words can lead to more acceptance mainly because they can increase the perceived teamwork, credibility, significance, and robustness in research evaluation. By revealing the unique strength of "We" in facilitating the recognition of new ideas in science, our study highlights the importance of small but mighty linguistic choices in promoting knowledge discoveries and breakthroughs.**

## Introduction

Linguistic writing is the primary channel through which the quality and merits of research ideas are communicated[1,2]. From grant proposals that secure vital funding to manuscripts that document groundbreaking discoveries, the linguistic choices researchers make can profoundly shape how their ideas and works are perceived and evaluated[3–6]. Yet, amid scarce attention and rigorous scrutiny of funding panels and peer reviewers, subtle elements of language are often overlooked in favor of methodological rigor and empirical novelty[7–10].

One small and often deemed unnoticeable linguistic marker in science writing is the use of first-person plural pronouns (i.e., "we," "us," "our," "ours"; referred to as "We" words hereinafter). Collective pronouns such as "We" may be particularly useful in an era of increasingly interdisciplinary research[11], where scientific advancements are largely driven by collaborative endeavor[12]. While the language of science has traditionally emphasized objectivity by avoiding personal pronouns[13,14], the use of "We" may signal credibility, robustness, collective effort, and shared accountability — qualities that can resonate with evaluators seeking to invest in impactful and reproducible work[15–19]. Yet, we lack a quantitative understanding of the predictive power of "We" in research evaluation and the psychocognitive mechanisms behind its potential power.

In this paper, we aim to investigate whether using more "We" words in scientific writing can lead to higher success rates in different research activities including science funding and publishing. We hypothesize that "We" words are powerful in obtaining favorable evaluation outcomes through several cognitive mechanisms. First, "We" creates a sense of shared identity between authors and reviewers (e.g., "These findings can advance progress in our community"), which can cause reviewers to perceive the work as more trustworthy or aligned with their own values[20]. Second, "We" signals teamwork, collaboration, consensus, and epistemic authority that add confidence to the conducted research[21,22]. This can increase perceived credibility and robustness. Third, "We" communicates modesty, politeness, and shared accountability and credit across a group[23,24]. Reviewers may perceive findings as less vulnerable to individual error or bias, which enhances likability and credibility. Fourth, "We" also softens self-promotion by depersonalizing achievements, which aligns with norms of humility in academia[25,26]. Reviewers may reward perceived humility with leniency, especially in disciplines where collaboration is prized.

To test these hypotheses, we systematically examined the statistical relationship between the

density of "We" words and high-stake review outcomes by utilizing three longitudinal and field-diverse datasets consisting of full samples of 13K grant proposals (both funded and unfunded) submitted to the Novo Nordisk Foundation (NNF, the world's wealthiest charitable foundation that supports science) and 47K paper manuscripts (both accepted and rejected) submitted to the BMJ Open (BMJO, a field-leading biomedical journal) and the International Conference on Learning Representations (ICLR, a top machine learning conference focused on LLM and AI).

Contrary to common expectation that overlooks style words, we find that grants and papers containing a higher frequency of "We" words are statistically and robustly associated with elevated success rates, even after controlling for important confounders including research topics, author impact, content novelty, and other linguistic factors. Furthermore, the effect of "We" is universally observed in various disciplines across different research activities including funding and publishing, and is applicable to both single-blind and double-blind review models.

To causally test the power of "We" in driving publishing success, we employed a large-scale LLM-based computational experiment where we substituted "We" words in each paper with demonstrative pronouns such as "this paper" and compared the change in ratings across two versions of all ICLR papers that were openly accessible. Importantly, our sophisticated experimental design enables us to uncover the cognitive mechanisms behind the power of "We" in research evaluation. Our findings shed fresh insights on how a small class of linguistic cues can have unexpected large impact on the success of ideas in the full scientific innovation pipeline.

## Data and Design

Our empirical datasets consist of (i) 13,520 grant applications submitted to NNF, (ii) 27,769 manuscripts submitted to BMJO, and (iii) 19,254 papers submitted to ICLR. NNF uses a central panel to evaluate grant proposals in medical, physical, and social sciences. BMJO employs the single-blind review model, consisting of both editors and external reviewers, whereas ICLR adopts the double-blind peer review model without the editorial process. In addition to the full-text content of grants and papers, we have access to PIs' and authors' demographic identities, bibliographic records indexed in the OpenAlex database[33], and detailed review comments. Our data possess unique strengths compared to prior studies. First, our datasets include both accepted and rejected samples, mitigating the selection bias in recent literature[3,8,34]. Second, our data have a broad and diverse scope as they include different research activities (both grants and papers) from

Table 1: Definition and measurement of all variables used in our regression models. A "NA" indicates that the variable is not applicable or absent in the corresponding dataset.

| Variable Names | Measurement | NNF | BMJO | ICLR |
|---|---|---|---|---|
| **Dependent Variables** | | | | |
| Acceptance | NNF funding decision is coded as 1 if proposal was funded and 0 otherwise. For BMJO and ICLR papers, this variable is coded as 1 for acceptance and 0 otherwise. The average NNF funding rate is 16.9%. The average acceptance rates are 37.9% for BMJO and 37.7% for ICLR papers. | **Y** | **Y** | **Y** |
| Reviewer rating score | The paper's rating score by a peer reviewer. To ensure consistency across years, all scores are normalized to the range [1, 10]. We lacked detailed review reports for NNF and BMJO datasets. | **NA** | **NA** | **Y** |
| Reviewer positivity | The positive sentiment score (range [-1, 1]) of a peer review report measured using TextBlob[27]. See details and robustness analyses in Methods. | **NA** | **NA** | **Y** |
| Reviewer engagement | The percentage of engagement words in a review report. See lexicon details in Methods. | **NA** | **NA** | **Y** |
| **Independent Variables** | | | | |
| Density of "We" words | The percentage of first-person plural pronouns (including "we," "us," "our," and "ours") in the full-text of grants or papers. The median density of "We" is 0.77% for NNF, 0.27% for BMJO, and 1.37% for ICLR. | **Y** | **Y** | **Y** |
| ***Grant/Paper content control*** | | | | |
| Number of words | The total number of words (log). The average length is 2,859; 6,877; 10,632 words for NNF, BMJO, ICLR. | **Y** | **Y** | **Y** |
| Title length | The total number of words in the title of a proposal or a paper. | **Y** | **Y** | **Y** |
| Density of promotional words | The percentage of promotional words in the full-text of grants or papers, measured based on a dictionary of 139 scientific promotional words[28]. See details in Methods. | **Y** | **Y** | **Y** |
| Readability | The reading score of full-text calculated using the Flesch Reading Ease formula [29]. | **Y** | **Y** | **Y** |
| Concreteness | The concreteness score computed based on the R package "doc2concrete"[30]. | **Y** | **Y** | **Y** |
| Number of references | The total number of papers referenced in a grant or paper. | **Y** | **Y** | **Y** |
| Novelty score | For NNF proposals, novelty is measured as the atypicality based on their reference combinations[31]. For BMJO and ICLR papers, novelty is computed as the semantic diversity of keywords. See details in Methods. | **Y** | **Y** | **Y** |
| Fixed-effects for submission year | The submission year of a proposal or a paper. | **Y** | **Y** | **Y** |
| Fixed-effects for research topics | There are three in-house research areas and five grant types identified by NNF proposals. For BMJO and ICLR papers, we used keyword embeddings to identify their main research topics. See details in Methods. | **Y** | **Y** | **Y** |
| ***PI/Author-level control*** | | | | |
| PI/Author gender | PI's gender was self-reported in NNF proposals. We name-inferred the corresponding authors' gender for BMJO papers using the GenderAPI[32]. Gender was not included for ICLR analyses due to its double-blind review process. We treated "Man" as the reference category. See details in Methods. | **Y** | **Y** | **NA** |
| PI/Author no. of prior publications | PI or corresponding author's number of publications up to the submission year. Author-level variables listed below were not included in ICLR data due to its double-blind review process. | **Y** | **Y** | **NA** |
| PI/Author no. of prior citations | PI or corresponding author's number of citations up to the submission year (log). | **Y** | **Y** | **NA** |
| PI no. of prior grant applications | The number of previous NNF grant proposals submitted by the PI at the time of each application. | **Y** | **NA** | **NA** |
| PI no. of prior funded grants | Same as above, but for the number of previously funded NNF grants. | **Y** | **NA** | **NA** |
| PI applied funding amount | The amount of funding applied for by the PI in each NNF proposal (log-transformed). | **Y** | **NA** | **NA** |
| PI age group | Self-reported by PI in NNF proposals. There are five age groups and the youngest group ([20, 29]) is used as the reference category. This variable is not available in the BMJO and the ICLR datasets. | **Y** | **NA** | **NA** |
| The number of authors | The number of authors in a paper. This variable is only applicable to the BMJO dataset. | **NA** | **Y** | **NA** |

a variety of disciplines (Medical, Physical, and Social Sciences, Biology, and Computer Science) encompassing academic venues that employ either single-blind or double-blind review model.

We employed fixed-effects regressions to examine the statistical correlation between "We" usage and the success of grant proposals and research papers (see details of all variables in Table 1). Our key independent variable is the percentage of "We" words in the full text of a proposal or a manuscript. The median density of "We" is 0.77% for NNF, 0.27% for BMJO, and 1.37% for ICLR. The variations of "We" usage highlight the need to test its effects across different disciplines.

We focused on three sets of dependent variables to examine our research questions: (i) grant funding or paper acceptance (binary), (ii) editorial decision and peer reviewer's recommendation for BMJO manuscripts, (iii) positivity and engagement of review comments for ICLR manuscripts. This design allows us to test the effect of "We" and its generalizability across different activities (grants and papers) by distinct evaluators (editors and peer reviewers) on various outcomes (final acceptance and depth of review comments). Our regression models include important control variables related to the full-text content (writing style, readability, novelty, etc.), PI or author characteristics (demographic identities, team size, prior productivity and citation impact, prior grant success and experience), and fixed effects for year, grant types, and research areas (see variable definitions and measurement in Table 1).

Finally, to establish causal evidence and investigate potential mechanisms behind the power of first-person plural pronouns, we leveraged the state-of-the-art LLM agent, GPT-4o, to conduct a large-scale computational experiment where we replaced "We" words in all ICLR abstracts with demonstrative pronouns (e.g., "our paper" vs. "this paper"). We then compared the LLM-evaluated rating scores between the original abstracts and the "we-substituted" replicas along five dimensions, including Decision Score and four important evaluation criteria (i.e., Credibility, Robustness, Teamwork, and Significance) that are designed to capture the hypothesized cognitive mechanisms behind the power of "We" words in research evaluation.

## Results

### More "We" Predicts Higher Success Rates of Grants and Papers

The median percentage of "We" in our three datasets is 0.77% for NNF proposals, 0.27% for BMJO papers, and 1.37% for ICLR papers, which translates to about one "We" word per 100

tokens. Would such a small class of function words have a sizable impact on an idea's success?

First, we find that the average percentage of "We" words is statistically higher in funded than in unfunded NNF proposals (0.97% vs. 0.84%, $p < 0.001$) and is also higher in accepted than rejected BMJO manuscripts (0.35% vs. 0.32%, $p < 0.001$). This pattern consistently replicates in accepted vs. rejected ICLR papers (1.43% vs. 1.38%, $p < 0.001$). Second, our logistic regressions show that the percentage of "We" is a strong and consistent predictor of success rates across all three datasets (see regression results in Supplementary Tables S1-S3), net of important control variables related to the full-text's content features (such as density of promotional words, readability score, degree of novelty, and applied funding amount), PI or author characteristics (such as age, gender, team size, prior success and experience, prior productivity and impact), and fixed-effects for application year and research areas (see variable definitions and measurements in Table 1).

Fig. 1 shows our models' marginal prediction of success rates as a function of the density of "We" words. For NNF grant applications, a one percentage point increase in the frequency of "We" is associated with about a 36.0% increase in the odds of funding ($\beta = 30.7, p < 0.001$, Supplementary Table S1), after taking into account the proposal's semantic features, degree of novelty, PI's self-reported gender and age, PI's prior experience and track records, and fixed effects for application year, research areas, and grant types. Fig. 1A indicates that the effect of "We" translates to more than a one-fold increase in the funding rate of NNF proposals from a low of 10.3% (95% CI = [8.5%, 12.1%]) to a high of 22.4% (95% CI = [18.5%, 26.3%]).

We observed a similar effect size of "We" words on paper acceptance. Specifically, for BMJO and ICLR papers, there is a 50.8% ($\beta = 41.1, p < 0.001$, Supplementary Table S2) and 37.2% ($\beta = 31.6, p < 0.001$, Supplementary Table S3) increase in the odds of acceptance for a one percentage point increase in the density of "We" contained in the full text paper. The marginal effects in Fig. 1B-C reveal that, across the full range of "We" in our data, the acceptance rate increases from 32.5% (95% CI = [30.3%, 34.6%]) to 42.0% (95% CI = [39.3%, 44.8%]) in BMJO papers and from 26.1% (95% CI = [23.1%, 29.0%]) to 47.6% (95% CI = [43.7%, 51.6%]) in ICLR papers, which again represents a nearly one-fold increase in the final acceptance rate.

Furthermore, we find that grants and papers with the median level of control features but are absent of sufficient "We" words experience a discount relative to the average level of success. Fig. 1A shows that proposals containing the median density of "We" words (0.77% in NNF) have a

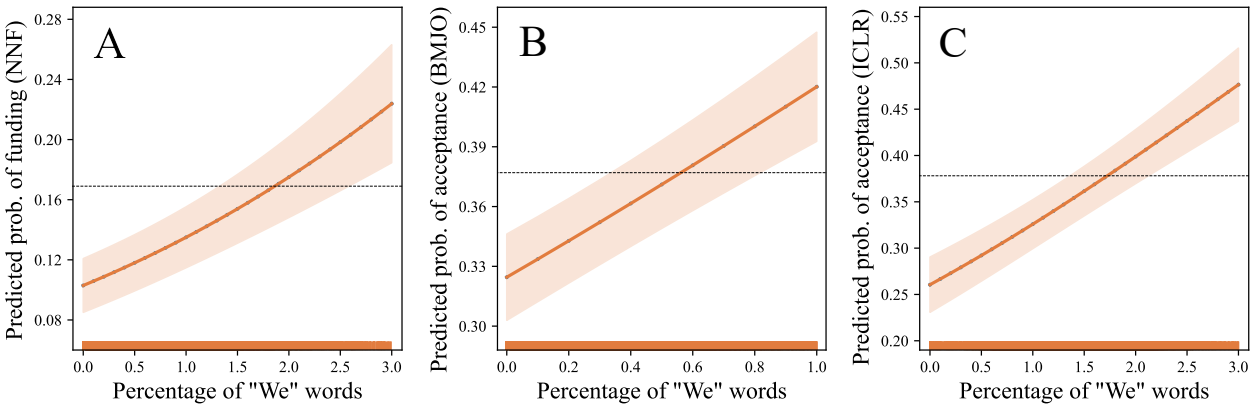


Figure 1: "We" words robustly predict the success rate of grant applications and paper manuscripts ($p < 0.001$ in all three regressions; see regression details in Supplementary Table S1-S3) after taking into account important confounding factors related to the content and PI or authors (see all variables in Table 1). Marginal plots show the predicted success rate as a function of the percentage of "We" words in (A) 13,520 NNF grant proposals, (B) 27,769 BMJO papers, and (C) 19,254 ICLR papers. The predictions are based on setting control variables at their median values. Error bars show the 95% bootstrapped confidence intervals. Dashed lines represent the average success rate in three datasets (16.9% for NNF, 37.7% for BMJO, 37.9% for ICLR, respectively). The bottom rug plots show the 99% (for NNF and ICLR) and 97% (for BMJO) density distribution of "we" words. Rerunning the same regressions by excluding observations above 3% (for NNF and ICLR) or above 1% (for BMJO) of "We" produced statistically similar results in each dataset ($p < 0.001$).

predicted funding probability 4 percentage points below the average funding rate (16.9% in NNF). Proposals that reach the average level of success are predicted to contain about 2.5 times the median density of "We" in NNF. The pattern is consistent for BMJO and ICLR papers with different peer review models: the predicted acceptance rate for papers with a median density of "We" is 2.8 percentage points (for BMJO) and 2.7 percentage points (for ICLR) below their average acceptance rates (Fig. 1B-C). To reach the average level of success, papers are predicted to contain 2.1 times (for BMJO) and 1.3 times (for ICLR) the median density of "We" words in the manuscript.

**The Power of "We" Applies to Both Editors and Reviewers in Single-Blind Review**

Evaluation in top journals often proceeds in several stages and multiple rounds involving different gatekeepers, including professional editors and external reviewers[8]. To examine the differen-

tial effects of "We" words on different actors, we leveraged the detailed review data of BMJO manuscripts. We focused on the first round of review, as evaluations in subsequent rounds were often dependent on previous rounds and reviewers' opinions were unlikely to be independent of each other after the first round due to social influence[35].

We ran a separate logit regression for two review stages. In the editorial-review stage, we treated each manuscript as an observation (N = 27,769) and coded the dependent variable as 1 if the manuscript was sent out for external review by editors (not being desk-rejected). In the peer-review stage, we treated each (manuscript, reviewer) pair in the first review round as an observation (N = 44,528) and coded the dependent variable as 1 if the manuscript received a minor revision or an acceptance recommendation from the reviewer. For both regressions, we included the same set of control variables as in Fig. 1B (see details in Table 1).

We find that the percentage of "We" used in BMJO manuscripts is statistically linked with a higher chance that the paper received a positive recommendation from both editors and reviewers. In particular, a one percentage point increase in the density of "We" can lead to a 53.5% ($\beta = 42.9, p < 0.001$, Supplementary Table S4) and a 12.1% ($\beta = 11.4, p = 0.002$, Supplementary Table S5) increase in the odds of obtaining a positive recommendation from editors and reviewers, respectively. Fig. 2 shows the marginal effects of "We" on positive decisions in each of the two review stages. It shows that, when the percentage of "We" increases from 0% to 1%, the positive decision rate of editors increases by 16.0% from 60.8% (95% CI = [58.5%, 63.2%]) to 70.5% (95% CI = [68.0%, 72.9%]) and reviewers' positive recommendation rate increases by 6.9% from 40.7% (95% CI = [38.8%, 42.6%]) to 43.5% (95% CI = [41.3%, 45.6%]).

This result demonstrates that the power of "We" applies to both professional editors and expert reviewers for a journal using the single-blind review. Fig. 2 also shows that "We" words have a much larger effect on editors than on reviewers. One possible reason might be related to the time difference they spend on evaluating the same manuscript. Research shows that decision-makers often lean on heuristics when cognitively challenging evaluations are hurried[36]. Editors typically make a desk-rejection decision in less than a week (faster decision), while reviewers can spend several months to thoroughly evaluate a paper (slower decision). Thus, compared to editors, reviewers may rely less on the paper's linguistic cues and focus more on its scientific content. Future research can extend our analysis to more journals and systematically test the underlying mechanisms behind the differential effects of "We" words on editors and reviewers.

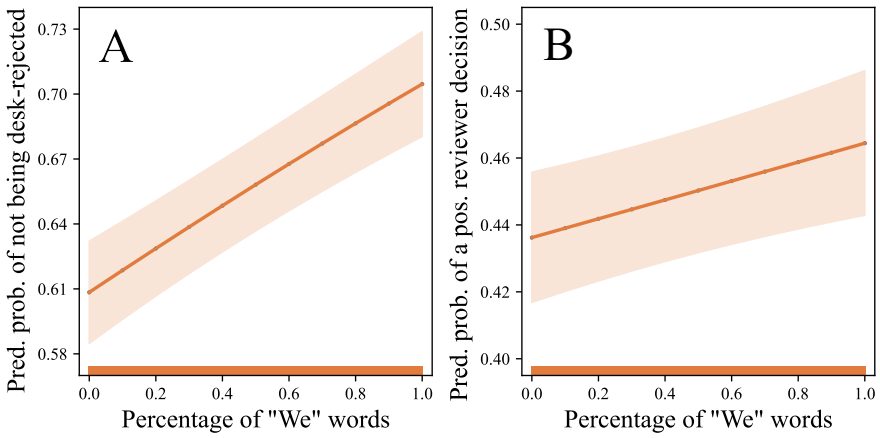


Figure 2: The density of "We" words is statistically associated with more positive recommendations by both editors and reviewers of BMJO manuscripts that used the single-blind review model. We applied the same regression specification as in Fig. 1B to predict (A) if a manuscript ($N = 27,769$) was not desk-rejected by editors ($p < 0.001$ in Supplementary Table S4) and (B) if a manuscript was recommended for direct acceptance or minor revision using $N = 44,528$ (manuscript, reviewer) pairs in the first round of review for 18,193 papers being sent out for peer review ($p = 0.002$ in Supplementary Table S5). Both plots show the predicted marginal effect of "We" words on positive decisions by setting control variables (see details in Table 1) at their median values. Error bars indicate the 95% bootstrapped confidence intervals. Rug plots show the 97% density distribution of "We" words in each setting. Running the same regressions by excluding data above 1% density of "We" produced statistically similar results in each setting ($p < 0.001$).

**"We" Increases Positive and Engaging Reviewer Feedback in Double-Blind Review**

To gain a deeper understanding of how "We" words can influence reviewers' decision-making process, we used the detailed review comments of ICLR papers to examine the statistical associations between "We" words and several linguistic indicators of positive reviewer feedback. ICLR is a top computer science conference that uses the double-blind review model, where each manuscript was evaluated by several independent peer reviewers without knowing the author's information. Analyzing this dataset further allows us to isolate potentially unobservable author-level characteristics.

Specifically, we treated each (paper, reviewer) pair as an observation (N = 70,365) in an OLS regression that uses the percentage of "we" words contained in a paper to predict three dependent

variables based on the full-text review comment: (1) reviewer's rating score, (2) positive sentiment score, and (3) reviewer's engagement score (see computation details in Methods).

We find that a higher frequency of "We" is statistically associated with a higher rating score ($\beta = 26.2, p < 0.001$, Supplementary Table S6), more positive sentiment ($\beta = 3.4, p < 0.001$, Supplementary Table S7-S8), and more engaging feedback from the reviewer ($\beta = 2.6, p < 0.001$, Supplementary Table S9). This indicates that a one percentage point increase in the density of "We" is linked to a 0.26 increase in rating score (which is 4.4% of the median rating), a 0.03 increase in positive sentiment (which is 0.5% of the median positivity), and a 0.03 increase in reviewer engagement (which is 1.4% of the median engagement).

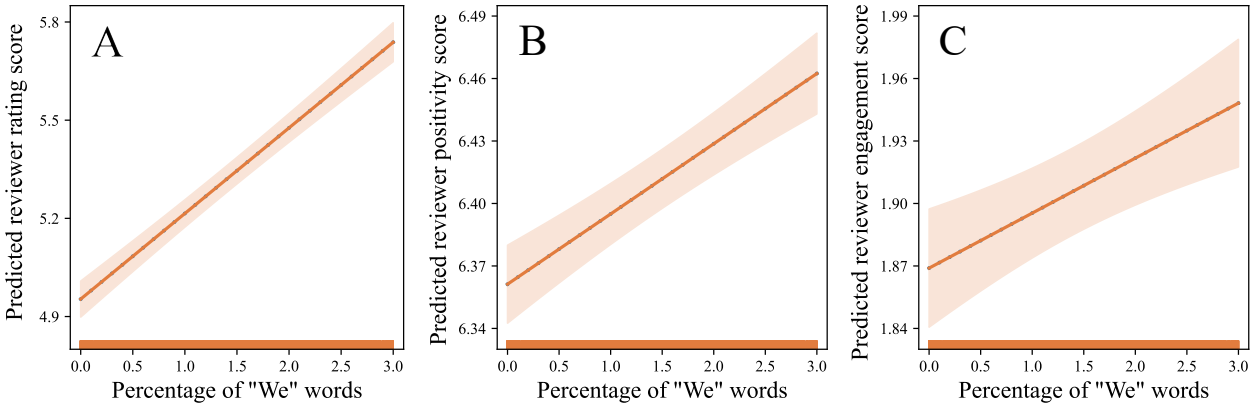


Figure 3: The density of "We" robustly predicts more positive and engaging reviewer feedback for ICLR papers that used the double-blind review model ($p < 0.001$ in Supplementary Table S6-S9). We applied an OLS model with the same specification as in Fig. 1C to predict the reviewer's (A) rating score, (B) positive sentiment, and (C) engagement level for $N = 70,365$ (paper, reviewer) pairs for all ICLR papers in our data. The predictions are based on setting control variables (see details in Table 1) at their median values. Error bars show the 95% bootstrapped confidence intervals. Rug plots show the 99% density distribution of "We" words. Running the same regression by excluding data above 3% of "We" produced statistically similar results ($p < 0.001$ in each setting).

The marginal plot in Fig. 3A shows that, when the percentage of "We" words increases from 0% to 3% in a ICLR paper, the predicted reviewer rating score increases by 16.0% from 4.95 (95% CI = [4.90, 5.01]) to 5.74 (95% CI = [5.68, 5.80]). Similarly, overall the full density range of "We" words, the predicted reviewer positivity (Fig. 3B) increases by 1.6% from 6.36 (95% CI = [6.34, 6.38]) to 6.46 (95% CI = [6.44, 6.48]) and the predicted reviewer engagement score (Fig. 3C)

increases by 4.3% from 1.87 (95% CI = [1.84, 1.90) to 1.95 (95% CI = [1.92, 1.98]).

These results demonstrate that a higher usage of "We" words used in ICLR papers is statistically linked with more positive and more engaging reviewer feedback, suggesting that "We" words may be able to increase the acceptance rate by generating more positive impressions of a paper's quality among reviewers. This finding motivates us to explore the underlying cognitive and psychological mechanisms that can explain the power of "We" in research evaluation.

**Causal Effect of "We" on Paper Acceptance and Four Cognitive Mechanisms**

The gold standard for testing causality in peer review is to recruit human experts to evaluate paper samples[37,38]. However, this method has two major limitations. First, it is challenging and costly to recruit enough participants. In practice, studies have to rely on crowdworkers who often do not possess domain expertise needed to assess a paper's merits[3]. Second, human-based behavioral experiments can employ only a few stimuli due to scalability limitation[3], which can cause generalizability concerns to subsequent findings. The recent advancements on LLM agents provide a good alternative to human participants, as they have been shown to derive human-like conceptual understanding and can outperform humans on many linguistic evaluation tasks[39–43].

To test the causal effect of "We" usage on paper acceptance, we employed GPT-4o, the state-of-the-art LLM with demonstrated human-level capabilities in science evaluation contexts[41,44], to conduct a large-scale controlled experiment using all ICLR papers. We focused on the ICLR dataset due to its public availability and a lack of permission to query public LLMs using the private NNF and BMJO data. In our experiment, we replaced "We" words in each abstract (N=19,254) with demonstrative pronouns such as "This paper" and instructed GPT-4o to evaluate both versions of each abstract on five dimensions, including Credibility, Teamwork, Significance, Robustness, and Decision Score (see definitions and the full prompt in Methods).

We used the paper's abstract instead of full text for two reasons. First, abstract captures the core hypotheses and findings of a paper while remaining concise and sufficiently self-contained for evaluation purposes. The median length of all ICLR abstracts is 181 words, which can provide essential information to reflect main findings, yet are short enough to be processed efficiently by both human raters and LLMs. Second, full-text evaluation would impose a significant time cost for LLMs at the scale of our ICLR dataset (N = 19,254) and can risk exceeding the token limitations and degrading model performance due to attention dilution[45], position bias[46–48], and long chain-

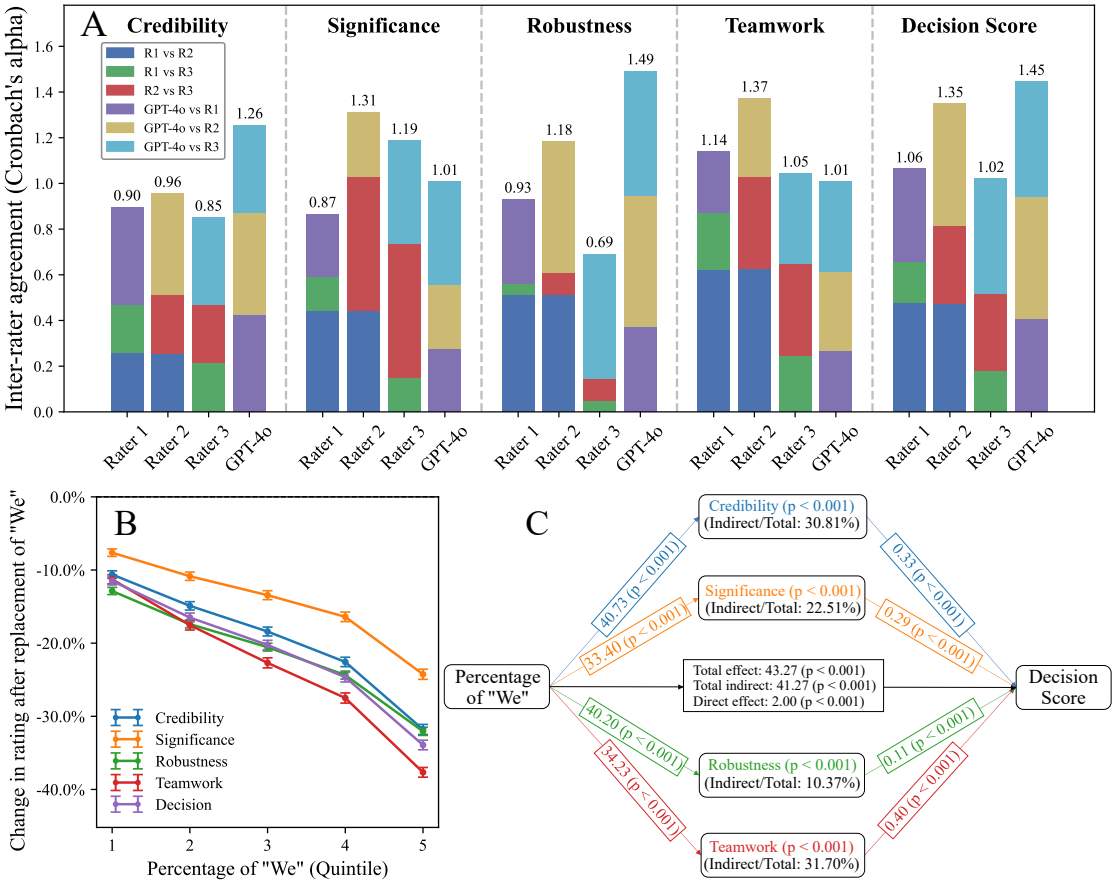


Figure 4: (A) The total pairwise agreement between each rater and the other three agents. Agreement is measured using Cronbach's alpha based on a random sample of 100 ICLR abstracts. GPT-4o shows higher and more stable agreement with humans than the pairwise agreement among three humans, especially on Credibility, Robustness, and Decision Score. (B) The Decision Score rated by GPT-4o decreases by 21.3% on average after replacing "We" with demonstrative pronouns (e.g., "this paper") in all 19,254 ICLR abstracts. The size of decline grows as the density of "We" increases from the bottom to the top quintile. Similarly, all four cognitive factors experience a significant decrease on average: 14.5% for Significance, 19.6% for Credibility, 21.4% for Robustness, 23.3% for Teamwork. Error bars indicate 95% confidence intervals based on standard errors. (C) Causal mediation analysis shows that Teamwork, Credibility, Significance, and Robustness indirectly accounts for 31.7%, 30.8%, 22.5%, and 10.4% of the total effect of "We" words on Decision Score, which collectively explain 95.4% of the power of "We" on acceptance. The analysis further controls for abstract length, readability, concreteness, and density of promotional words (Table 1).

of-thought bias[49]. Abstract-level analysis thus minimizes token overflow and preserves attention stability across samples and is more practical in our large-scale experiment.

To further justify our use of LLM-based evaluations and establish reliability of GPT-4o's evaluation capabilities, we performed the Alternative Annotator Test[44] by comparing GPT-4o with human experts based on a random sample of 100 ICLR abstracts (see details on human recruitment and training in Methods). Fig. 4A shows the pairwise agreement score (measured using Cronbach's alpha) per agent on five cognitive dimensions. We find that GPT-4o's ratings are more consistent than three human raters, especially on Credibility, Robustness, and Decision Score (Fig. 4A). In fact, the average agreement between GPT-4o and three human raters is either higher or more stable than the average pairwise agreement among three raters for all five dimensions: Credibility (mean: 0.42 vs 0.24, std: 0.03 vs 0.02), Significance (mean: 0.34 vs 0.39, std: 0.10 vs 0.22), Robustness (mean: 0.50 vs 0.22, std: 0.11 vs 0.26), Teamwork (mean: 0.34 vs 0.43, std: 0.06 vs 0.20), Decision Score (mean: 0.48 vs 0.33, std: 0.07 vs 0.15). Besides better performance, GPT-4o is also 30 times faster and three orders of magnitude cheaper than human raters (see details in Methods). These advantages demonstrate that LLM agents are capable of evaluating academic papers and are more suitable at the scale of our experiment because of their efficiency and scalability.

We then prompted GPT-4o to evaluate all 19,254 ICLR abstracts and their replicas after the replacement of "We". It takes GPT-4o agents less than 4 days to evaluate all abstracts and their "without-We" replicas, which would take a human rater about 140 days to complete the same task. The total cost of GPT-4o is about $30 USD across preliminary model selection, prompt iteration, and final assessment of all abstracts. Our study design thus provides future research a practical framework for conducting experiments on large corpus of scientific texts. Our detailed prompts, rating guidelines, and instruction protocols are provided in the Methods section under LLM-based Computational Experiment.

We compared the ratings of two versions of the same abstract along five dimensions. First, we find that GPT-4o's Decision Score decreases significantly after replacing "We" words in ICLR abstracts with demonstrative pronouns (Fig. 4B). Second, the size of decline grows sharply as the density of "We" increases: from the bottom to the top quintile of "We" words' density, the relative decrease in Decision Score grows from 11.5% (95% CI = [10.9%, 12.0%]) to 33.9% (95% CI = [33.3%, 34.6%]). This result provides robust causal evidence of "We" on paper acceptance.

Fig. 4B also shows a growing and consistent decline in GPT-4o's rating scores after the replacement of "We" words across all four cognitive dimensions, including Teamwork, Credibility, Robustness, and Significance. On average, the rating score decreases by 14.5% for Significance, 19.6% for Credibility, 21.4% for Robustness, and 23.3% for Teamwork after the replacement of "We" words (Fig. 4B). This result suggests that these four conceptual factors may partly or fully explain the causal impact of "We" on paper acceptance. To test these cognitive mechanisms, we performed a causal mediation analysis using our experimental data where we further controlled for four linguistic variables of each abstract, including readability, concreteness, number of words, and percentage of promotional words (see variable definitions and measurements in Table 1).

Fig. 4C shows that a higher density of "We" words indeed leads GPT-4o to perceive the paper abstracts as more collaborative ($\beta$ = 34.2, $p < 0.001$), more trustworthy ($\beta$ = 40.7, $p < 0.001$), more significant ($\beta$ = 33.4, $p < 0.001$), and more robust ($\beta$ = 40.2, $p < 0.001$), whose indirect effects account for 31.7% (effect size = 13.7, $p < 0.001$), 30.8% (effect size = 13.3, $p < 0.001$), 22.5% (effect size = 9.7, $p < 0.001$), and 10.4% (effect size = 4.5, $p < 0.001$) of the total effect of "We" (effect size = 43.3, $p < 0.001$) on Decision Score, respectively. Crucially, the four factors collectively can explain about 95.4% ($p < 0.001$) of the total impact of "We" on acceptance decision (Fig. 4C), suggesting that they are the main cognitive mechanisms through which "We" impacts research evaluation. Taken together, these results support our hypothesis that a higher usage of "We" can lead to a more positive evaluation outcome because "We" words can improve the perceived teamwork, credibility, significance, and robustness of a paper beyond its content.

## Discussion

Rhetoric is a crucial element of scientific innovation. While science is often presented as purely objective, rhetorical strategies are essential for developing, establishing, and disseminating new ideas within the scientific community and the broader public[50,51]. Yet, we know little about how scientists navigate the intricacies of presenting the qualities of their ideas to funders and reviewers[4,52,53]. Further, the increasing methodological complexity and the growing interest in interdisciplinary research[11] pose challenges for researchers to effectively convey the merits of fresh ideas[54,55]. Motivated by communication literature that identifies personal pronouns as powerful tools for shaping persuasion and interpersonal dynamics, here we examined the effect of first-person plural pronouns ("We" words) on the recognition of scientific ideas in different evaluation contexts.

Our empirical design leverages three diverse datasets encompassing 13K grant proposals submitted to the Novo Nordisk Foundation and 47K papers submitted to BMJO and ICLR from 2011 to 2024. We find a universal effect of "We" words on success across research activities. Specifically, a one percentage point increase in "We" words is linked with a 36-51% increase in the odds of success, even after accounting for factors such as research areas, content novelty, writing styles, and author impact. Furthermore, the effect of "We" persists across single-blind (BMJO) and double-blind (ICLR) review processes, influencing both editors and reviewers, and is linked to more positive and engaging reviewer feedback. Our findings demonstrate that the strategic use of "We" is associated with higher success rates in both science funding and publication.

Critically, our LLM-assisted computational experiment that replaces "We" words with demonstrative pronouns provides causal evidence on the power of "We" in acceptance. Digging into the underlying mechanisms, we find that "We" words drive more success because these collective pronouns can enhance the perceived teamwork, credibility, significance, and robustness of papers. Our findings underscore the pivotal role of subtle rhetorical strategies in the gatekeeping of scientific knowledge, revealing how small linguistic patterns can have unexpectedly large impact on human perceptions and can amplify the recognition of new ideas in science.

Our study has several important implications for research on innovation. First, we analyzed diverse datasets containing the full text of both rejected and accepted projects across domains, including BMJO papers in biomedicine, ICLR papers in machine learning, and grants from medical, physical, and social sciences. Our design demonstrates how rich empirical datasets combined with modern statistical models and computational linguistic techniques can lead to robust and generalizable findings on the power of language in facilitating scientific innovation, which we hope can serve as an inspiration for future research.

Second, our findings reveal that linguistic style is as important as content for the recognition of meritorious ideas in innovation ecosystems. Understanding the importance of discourse may facilitate knowledge production and improve equality in science[56,57]. Prior research shows that groundbreaking ideas often face resistance as they threat old paradigm[58–60] and that the degree of resistance is correlated with author demographics[57,61]. Our research suggests that "We" might help overcome inertia to bold ideas by fostering a sense of shared ownership and accountability, thereby reducing the perceived risk among funders and reviewers. Our findings also underscore the need to reevaluate how linguistic choices used by researchers from diverse backgrounds (e.g., scholars

who use English as the first vs. second language) intersect with inequalities in science[62,63].

Third, our LLM-assisted experiment exemplifies the transformative potential of AI in scientific inquiry, aligning with the emerging "AI for science" paradigm[64,65]. By employing LLM agents to evaluate counterfactual papers, our work provides a scalable framework for causal analysis in observational data, addressing endogeneity challenges commonly seen in the science of science literature[28,66,67]. Similarly, building on recent AI applications in automated literature reviews and hypothesis generation[68], future research can utilize LLMs to directly generate counterfactual texts while preserving semantics to examine other linguistic markers of success. On the other aspect, as AI tools become integral to research assistance, science-specific AI systems could incorporate "We" optimization in scientific writing aids. However, this also raises ethical considerations: if access to advanced AI systems is uneven, such practice could exacerbate existing inequalities.

The policy ramifications of our study are multifaceted, offering actionable insights for funding agencies, journals, and academic institutions. In a landscape where funding is competitive and paper rejection is the norm, understanding these subtle linguistic signals could inform training for early-career researchers, refine evaluation processes to mitigate unconscious biases, and promote more equitable scientific discourse. First, training programs for junior scholars should emphasize the importance of rhetorical strategies, including "We" word usage, to level the playing field in grant and publication competitions, potentially through workshops integrated into PhD curricula or funding agency guidelines. Second, peer-review reforms could incorporate linguistic audits to mitigate cognitive biases and misjudgment, such as training reviewers to recognize how pronouns influence perceptions, fostering more equitable and objective evaluations. Third, AI-assisted tools for proposal drafting and paper editing, when endorsed by funders and publishers, could embed "We" recommendations to boost success rates, particularly for underrepresented groups[52,66]. Finally, on a broader scale, these findings advocate for interdisciplinary policies that recognize communication and presentation style as a core competency in science practice[54].

In conclusion, by elucidating the causal power of "We" words in the success of scientific ideas in funding and publications, our study may change existing practice in science that overtly emphasizes content and novelty over discourse, which can open up new directions aimed at understanding and promoting innovation from the linguistic perspective. Future research could explore cross-cultural variations in pronoun effects and extend our framework to identify other linguistic features to further uncover the rhetorical strength of discovery.

## Methods

### Ethics Statement

This study is fully reviewed and approved by the Research Office of the City University of Hong Kong (HU-STA-00001180) and complies with all pertinent ethical regulations. The informed consent requirement was determined to be waived regarding our analyses of grant applications and paper manuscripts submitted to NNF, BMJO, and ICLR.

### Grants and Paper Peer Review Data

Our datasets contain full samples of the full text of both accepted and rejected grant applications and paper manuscripts from major academic disciplines including Medical Sciences, Physical Sciences, Social Sciences, Biology, AI, Machine Learning, and Computer Science.

**NNF grant proposals.** The NNF dataset contains 13,520 grant proposals submitted between 2015 and 2022 to the Novo Nordisk Foundation (NNF), one of the largest science funders in the world. There are three research areas (including Medical, Physical, and Social Sciences) and five grant types (such as Research Projects and Investigator Grants) among all NNF grant proposals. Each proposal contains rich metadata, including the funding decision, full text of project description, application year, the program area and grant type, the applied funding amount, as well as deidentified PI personal information, including PI's self-reported gender, age, publication and citation records. We also calculated PI's number of prior applications and number of prior funded NNF grants. The average proposal length is 2,859 words. The average funding rate is 16.9%.

**BMJO paper submissions.** This dataset contains 27,769 rejected and accepted manuscripts submitted to the journal *BMJ Open* (BMJO) between 2011 and 2022. BMJO is one of field-leading journals that publish all topics related to medical research. It employs the single-blind review model, which means that author identities are accessible to both editors and reviewers. In addition to the full text of each manuscript and its review comments, we identified its corresponding authors in the OpenAlex database[33] based on ORCID and obtained their publication and citation records. The average manuscript length is 6,877 words. The desk-rejection rate is 33.7%. The final acceptance rate is 37.7%. There are 44,528 (manuscript, reviewer) pairs in total for manuscripts that were sent out for peer review by editors in the first review round.

**ICLR paper submissions.** We collected the full text of both accepted and rejected re-

search papers (N = 19,254) submitted to the *International Conference on Learning Representations* (ICLR) between 2018 and 2024 from `https://openreview.net/`. Unlike BMJO, the ICLR conference does not have an editorial review process and employs only one round of peer review. Each paper was evaluated in a double-blind process by 3-4 independent reviewers, which means that author characteristics had a limited impact on review outcomes and were thus excluded in our analysis. There are 70,365 (paper, reviewer) pairs, with each containing a rating score and detailed review reports. The number of ICLR submissions increased sharply from 909 in 2018 to 5,771 in 2024 in our data. On average, a paper contains 10,638 words, and a review report contains 477 words. We treated manuscripts labeled as Oral, Spotlight, and Poster as accepted papers. The average acceptance rate is 37.9%. Note that we collected the ICLR dataset before the OpenReview Identity Leak accident, thus reviewer information was excluded in our analysis.

**Regression Models**

We used fixed-effects regression models to examine the statistical relationship between the percentage of "We" words and various dataset-specific review outcomes. For binary dependent variables, we used logit regressions. For numeric dependent variables, we used linear regressions. Our models take into account various important confounders related to the grant's (or paper's) semantic content, the PI's (or author's) individual characteristics, and fixed-effects for the submission year and research fields (See details of all variables and their availability by dataset in Table 1).

**Linguistic Measures**

We used standard NLP methods and Python/R packages to compute the following linguistic variables for our three submission datasets and the full-text review comments of ICLR papers.

**Density of "We" words.** This is our key independent variable. We counted the percentage of first-person plural pronouns, including "we", "us", "our", and "ours", in the full-text of each submitted document (lowercased). The median density of "We" words is 0.77% in NNF proposals, 0.27% in BMJO manuscripts, and 1.37% in ICLR papers.

**Title and full-text length.** We counted the total number of words in each document and its title after standard text cleaning such as lowercase and removing punctuations. To reduce the impact of outliers, we log-transformed the total number of words in the full-text.

**Density of promotional words.** We used a lexicon of 139 scientific promotional words de-

veloped in the literature[69] and further validated in recent studies[4,28]. Some exemplary promotional words include "crucial," "unique," and "substantial" (see the full dictionary in Supplementary Note A). These promotional words have been found to hold great strength in communicating the merits of innovative ideas in grant applications[28]. We calculated the percentage of these promotional words in the full-text of each submitted document.

**Readability.** Readability was assessed using the Python package NLTK (Natural Language Toolkit) based on the Flesch Reading Ease score[29]. This measure combines the average sentence length and the average syllables per word to estimate the textual complexity. A higher readability score indicates an easier-to-read text, whereas a lower score corresponds to a text with greater syntactic and lexical complexity.

**Concreteness.** Lexical concreteness was measured using the R package "doc2concrete" that leverages a dictionary of word-level concreteness rating scores developed based on standardized psycholinguistic norms[30]. This method outputs the concreteness score averaged over all words contained in each document. A higher score indicates a more grounded and concrete writing style.

**Reviewer positivity.** To measure the degree of positive sentiment contained in the full-text of review comments, we computed the positivity score using the Python TextBlob package[27]. This package uses the lexicon-based approach to measure positive sentiment. A higher score implies a more positive reviewer feedback. To make the results comparable to those of reviewer rating scores, we normalized the positivity scores to the same range [1, 10]. We obtained consistent results when using SciBERT to measure positive sentiment (Supplementary Table S8). SciBERT utilizes context-aware word embeddings specifically trained on large corpus of scientific text, which is suitable in science-specific sentiment predictions tasks[70].

**Reviewer engagement.** We calculated the percentage of engagement words in the full text of review comments. We used the lexicon of 116 engagement words developed in a previous work[3] based on a large corpus of peer-review reports published by Nature Communications. These words are indicative of reviewer's responsiveness in providing engaging feedback to authors, such as "address," "revise," "concern," "response," "improve," "version," "recommend" (see the full dictionary in Supplementary Note B). Similar to sentiment positivity, we normalized the engagement scores to the range [1, 10] in order to compare with the analysis of reviewers' rating scores.

**Gender Labels**

We used the PI's self-reported binary gender labels for all NNF proposals (65.68% from Men and 34.32% from Women). The gender variable was not included in our analyses of ICLR papers because the author information was unavailable to reviewers in the double-blind review setting.

For BMJO papers, we lacked information on how authors self-identify themselves. We thus inferred the gender for all corresponding authors based on their author names using the Gender API [32], which outputs a binary gender label with an associated probability. We used a 0.75 threshold to obtain a distinct binary gender label for each corresponding author (otherwise labeled as "Unisex"). As a result, 41.8%, 31.9%, and 26.3% of papers are classified as belonging to Men, Women, and Unisex. We validated the classifier's prediction accuracy using a random sample of 1,000 authors with gender labels manually verified based on their official profile photos, Wikipedia entries, or Google searches. The classifier achieves a F1 score of 0.94 for Man (N = 358) and 0.91 for Woman (N = 254). Using different thresholds produced statistically similar results.

**Novelty Measures**

For NNF proposals, we computed their novelty scores using a popular and validated measure that characterizes a grant's innovativeness based on the degree to which it combines past literature in either familiar or novel ways[31]. This measure operates at the level of journal combinations in a proposal's cited references. For a given journal pair at a given year, the measure quantifies its novelty score by comparing its observed co-citation frequency in all papers published in that year with the frequency expected by chance in a randomized citation network that shuffles citation links while preserving the yearly citation statistics for each paper. A novelty score below 0 implies a less common and more novel journal combination, whereas a score above 0 indicates a more familiar and conventional combination. The method outputs a distribution of novelty scores for a given proposal (one score per journal pair). Following a previous study[28], we used the median of all negative scores as the degree of novelty for each proposal (reversed coded for easier interpretation). For a small fraction of proposals (less than 7%) that lacked the detailed reference data needed for novelty calculation, we estimated its novelty as the average score of 10 most similar proposals based on semantic content. The similarities between all proposals were quantified as the cosine similarity of vector embeddings learned from their brief project descriptions.

For both BMJO and ICLR papers, we computed their novelty scores based on their keyword embeddings obtained using the model "Specter" developed by the Allen Institute for AI[71]. We used

this keyword embedding framework instead of the journal combination approach for two considerations: (i) the novelty scores produced by two approaches tend to be highly correlated[72], and (ii) the embedding-based method is computationally more efficient and does not suffer from noise information in a paper's references. We calculated the pairwise cosine distance for all keywords in the embedding space for each paper, and used the 90th percentile cosine distance as the paper's summary novelty score. Papers with higher (lower) novelty scores tend to combine more distant (similar) keywords in the knowledge space.

**Research Topics**

NNF proposals are classified into three major research areas identified by NNF's in-house panel experts, including Medical, Physical, and Social Sciences. We identified research topics of BMJO and ICLR papers based on their keyword embeddings obtained in the previous section. Specifically, we clustered all 21,211 unique keywords used in ICLR papers into 7 topics and all 18,411 unique keywords used in BMJO papers into 14 topics. The number of optimal clusters was identified based on the Silhouette Score[73], a standard metric commonly used to measure cluster quality. Varying the number of topic clusters produced statistically similar results for both datasets. In our analysis, each paper can belong to multiple research topics depending on its keyword-topic membership in our clustering.

**LLM-based Computational Experiment**

**Experimental design.** We instructed GPT-4o to simulate the role of an experienced AI researcher and asked the LLM agent to evaluate each paper's abstract with the following prompt:

> You are an experienced researcher specialized in machine learning, NLP, and AI. Now you are invited to review a paper submitted to a top AI conference. Please evaluate the following paper abstract in depth across five dimensions, using your expert reasoning with domain knowledge:
>
> - Credibility: How credible do you think the paper is? Credibility in research evaluation refers to the trustworthiness, validity, and reliability of research findings, methods, and interpretations. Higher scores indicate a more credible paper.
> - Significance: How significant is the problem this research aims to address? Significance in research refers to the importance, impact, and contribution of a study to its field, society, or broader scientific knowledge. Higher scores suggest that

the paper is more significant.

- Robustness: How robust does the finding of this paper sound to you? Robustness in research evaluation refers to the reliability, reproducibility, and generalizability of findings under varying conditions, methodologies, and datasets. Higher scores reflect stronger methodological rigor.
- Teamwork: To what extent does this paper reflect a sense of teamwork? Teamwork implies that the research is conducted by a group of people. Higher scores mean that this work is more likely to be the product of a collaborative effort.
- Decision: How likely would you recommend acceptance of this paper? This score should not be an average of the previous ones. Instead, it should reflect your independent judgment. Make this final decision through holistic reasoning.

Please think and reflect deeply before assigning a score between 1 and 10. Your final output should follow this format: "Credibility: [score]", "Significance: [score]", "Robustness: [score]", "Teamwork: [score]", "Decision: [score]".

Each request consisted of a single abstract accompanied by a prompt-based evaluation instruction listed above. To reduce time latency and computational cost, we interacted with GPT-4o through OpenAI's batch API to process multiple abstracts asynchronously in parallel. All prompts were issued in a zero-shot setting with no in-context examples provided, as recent studies have shown that LLM can function as reliable zero-shot evaluators in various social, linguistic, and psychological annotation tasks[74]. We also instructed GPT-4o to perform chain-of-thought reasoning to improve accuracy and performance[75].

**Comparison between GPT-4o and human raters.** We compared GPT-4o against three human raters based on a random sample of 100 ICLR abstracts. Rater 1 is an undergraduate student majoring in Statistics and Data Science. Rater 2 is a graduate student majoring in Data Science. Rater 3 holds a bachelor's degree in Computer Science and a master's degree in Information Science. All three raters come from universities that use English as the medium of instruction. They received identical written instructions on our project goal, task description, and evaluation criteria. (see details in Supplementary Note C)

Each rater was asked to play the role of an experienced researcher in machine learning, natural language processing, and artificial intelligence, and to review abstracts of papers submitted

to a top-tier AI conference along five dimensions, including Credibility, Teamwork, Significance, Robustness, and Decision Score. Each rater underwent a pre-test on 20 abstracts and was trained by one of the authors to improve their understanding of the task and rating consistency. This process was iterated several times until three raters reached an acceptable agreement (Cronbach's alpha = 0.50). All three raters worked independently to complete their evaluation of 100 ICLR abstracts.

Evaluating a single abstract costs about 5 minutes for each human rater, and the time cost increases further when multiple abstracts are assessed consecutively. In contrast, GPT-4o takes only 5 seconds on average to evaluate each abstract, which is about 30 times faster. Furthermore, it takes about 10 hours per human rater for completing 100 abstracts, which is equivalent to a monetary cost of about $360 USD based on the average hourly wage in the U.S. in 2025. However, the cost of GPT-4o for the same task is less than 10 cents, which is about 3,600 times cheaper.

**Mediation Analysis**

We applied a standard causal mediation analysis framework to our experimental data to quantify the impact of four mediators in explaining the predictive power of "We" words on acceptance. The analysis consists of three OLS regression models. The total effect model predicts the Decision Score using the percentage of "We" words. The mediator model predicts the rating score of each mediator using the percentage of "We" words. The direct effect model predicts the Decision Score using both "We" words and four mediators. To further account for content variations between different observations, we added four linguistic controls, including abstract length, readability, concreteness, and density of promotional words in all three regression models.

**Data and Code Availability**

The deidentified NNF grant dataset is available upon request through an NDA process with the Novo Nordisk Foundation. The deidentified BMJO manuscript dataset is available upon request through an NDA process with the journal BMJ Open. The ICLR paper dataset is publicly available at `https://openreview.net/group?id=ICLR.cc`. The OpenAlex bibliometric database used in this study is openly accessible at `https://openalex.org/`. The code for replicating our results can be accessed on Github at `https://github.com/xyz`.

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# The Power of "We" in Science Funding and Publication (Supplemental Materials)

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## I. SUPPLEMENTARY TABLES

TABLE S1. Logistic regression models for predicting the likelihood of an NNF application being funded. Model 1 includes only the density of "We" words and fixed-effects controls. Model 2 adds content controls. Model 3 adds PI-related controls. Coefficient estimates are reported with $p$-values shown in parentheses. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 2$).

| | *Dependent variable: Funding Decision (Funded = 1)* | | |
|---|---|---|---|
| | **Model 1** | **Model 2** | **Model 3** |
| Perc. of "We" Words | 38.866 ($p < 0.001$) | 37.746 ($p < 0.001$) | 30.719 ($p < 0.001$) |
| Perc. of Promotional Words | | 36.520 ($p < 0.001$) | 37.891 ($p < 0.001$) |
| Flesch Reading Score | | −0.012 ($p = 0.555$) | −0.006 ($p = 0.769$) |
| Concreteness Score | | −0.080 ($p = 0.002$) | −0.079 ($p = 0.003$) |
| Novelty Score | | 0.002 ($p = 0.271$) | 0.001 ($p = 0.602$) |
| Number of Words in Title | | 0.121 ($p = 0.088$) | 0.064 ($p = 0.371$) |
| Number of Words in Full Text (log) | | 0.212 ($p < 0.001$) | 0.284 ($p < 0.001$) |
| Number of References | | 0.009 ($p < 0.001$) | 0.009 ($p < 0.001$) |
| PI Applied Funding Amount (log) | | | −0.052 ($p = 0.031$) |
| PI No. Prior Grant Applications | | | −0.038 ($p = 0.004$) |
| PI No. Prior Funded Grants | | | 0.351 ($p < 0.001$) |
| PI No. Prior Publications | | | −0.001 ($p = 0.194$) |
| PI No. Prior Citations (log) | | | 0.070 ($p < 0.001$) |
| PI Gender (2 categories, self-reported) | Y | Y | Y |
| PI Age (6 categories, self-reported) | Y | Y | Y |
| Application Year (2015–2022) | Y | Y | Y |
| Instrument Type (5 categories) | Y | Y | Y |
| Research Areas (3 categories) | Y | Y | Y |
| Intercept | −2.342 ($p < 0.001$) | −5.608 ($p < 0.001$) | −5.529 ($p < 0.001$) |
| Pseudo-$R^2$ | 0.022 | 0.029 | 0.045 |
| Observations | 13,520 | 13,520 | 13,520 |

TABLE S2. Logistic regression models for predicting the probability of a BMJO paper manuscript being accepted. Model 1 includes only the density of "We" words and fixed-effects controls. Model 2 adds content controls. Model 3 adds author controls. Coefficient estimates are reported with $p$-values shown in parentheses. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 1.5$).

| | *Dependent variable: Final Acceptance (Accept = 1)* | | |
|---|---|---|---|
| | **Model 1** | **Model 2** | **Model 3** |
| Percentage of "We" Words | 30.452 ($p < 0.001$) | 45.109 ($p < 0.001$) | 41.064 ($p < 0.001$) |
| Percentage of Promotional Words | | 11.846 ($p = 0.002$) | 14.207 ($p < 0.001$) |
| Flesch Reading Score | | −0.255 ($p < 0.001$) | −0.223 ($p < 0.001$) |
| Concreteness Score | | 0.158 ($p < 0.001$) | 0.176 ($p < 0.001$) |
| Novelty Score | | −0.107 ($p = 0.586$) | −0.145 ($p = 0.460$) |
| Number of Words in Title | | 0.029 ($p < 0.001$) | 0.024 ($p < 0.001$) |
| Number of Words in Full Text (log) | | 0.750 ($p < 0.001$) | 0.665 ($p < 0.001$) |
| Number of References | | 0.005 ($p < 0.001$) | 0.005 ($p < 0.001$) |
| Number of Authors | | | 0.041 ($p < 0.001$) |
| Corr. Author No. Prior Publications | | | 0.000 ($p < 0.001$) |
| Corr. Author No. Prior Citations (log) | | | 0.032 ($p < 0.001$) |
| Corr. Author Gender (3 categories) | Y | Y | Y |
| Submission Year (2011–2022) | Y | Y | Y |
| Research Topics (14 categories) | Y | Y | Y |
| Intercept | 0.040 ($p = 0.779$) | −7.116 ($p < 0.001$) | −6.685 ($p < 0.001$) |
| Pseudo-$R^2$ | 0.060 | 0.093 | 0.100 |
| Observations | 27,769 | 27,769 | 27,769 |

TABLE S3. Logistic regression models for predicting the probability of an ICLR paper being accepted. Model 1 includes only the density of "We" words and fixed-effects controls. Model 2 adds content controls. Author-related controls are not considered due to ICLR's double-blind review process. Coefficient estimates are reported with $p$-values shown in parentheses. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 1.5$).

| | *Dependent variable: Final Acceptance (Accept = 1)* | |
|---|---|---|
| | **Model 1** | **Model 2** |
| Percentage of "We" Words | $21.664\,(p < 0.001)$ | $31.638\,(p < 0.001)$ |
| Percentage of Promotional Words | | $34.947\,(p < 0.001)$ |
| Flesch Reading Score | | $-0.082\,(p < 0.001)$ |
| Concreteness Score | | $-0.100\,(p < 0.001)$ |
| Novelty Score | | $0.878\,(p = 0.004)$ |
| Number of Words in Title | | $-0.018\,(p = 0.003)$ |
| Number of Words in Full Text (log) | | $1.963\,(p < 0.001)$ |
| Number of References | | $0.003\,(p < 0.001)$ |
| Submission Year (2018–2024) | Y | Y |
| Research Topics (7 categories) | Y | Y |
| Intercept | $-0.883\,(p < 0.001)$ | $-18.816\,(p < 0.001)$ |
| Pseudo-$R^2$ | 0.008 | 0.087 |
| Observations | 19,254 | 19,254 |

TABLE S4. Logistic regression model for predicting the probability of a BMJO paper not being desk-rejected by editors. Coefficient estimates are reported with $p$-values shown in parentheses. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 1.5$).

| | *Dependent variable:* *Editorial Acceptance (No desk-rejection = 1)* |
|---|---|
| Percentage of "We" Words | 42.861 ($p < 0.001$) |
| Percentage of Promotional Words | −0.951 ($p = 0.798$) |
| Flesch Reading Score | −0.167 ($p < 0.001$) |
| Concreteness Score | 0.198 ($p < 0.001$) |
| Novelty Score | −0.059 ($p = 0.764$) |
| Number of Words in Title | 0.017 ($p < 0.001$) |
| Number of Words in Full Text (log) | 0.796 ($p < 0.001$) |
| Number of References | 0.006 ($p < 0.001$) |
| Number of Authors | 0.009 ($p = 0.004$) |
| Corr. Author No. Prior Publications | 0.000 ($p < 0.001$) |
| Corr. Author No. Prior Citations (log) | 0.009 ($p = 0.155$) |
| Corr. Author Gender (3 categories) | Y |
| Submission Year (2011–2022) | Y |
| Research Topics (14 categories) | Y |
| Intercept | −5.804 ($p < 0.001$) |
| Pseudo-$R^2$ | 0.093 |
| Observations | 27,769 |

TABLE S5. Logistic regression model for predicting the probability of a BMJO paper receiving positive reviewer recommendations. Each (paper, reviewer) pair is used as an observation. Coefficient estimates are reported with $p$-values shown in parentheses. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 1.5$).

| | *Dependent variable:* *Reviewer Rec. (Accept & Minor Rev. = 1)* |
|---|---|
| Percentage of "We" Words | 11.399 ($p = 0.002$) |
| Percentage of Promotional Words | 26.703 ($p < 0.001$) |
| Flesch Reading Score | −0.172 ($p < 0.001$) |
| Concreteness Score | 0.018 ($p = 0.105$) |
| Novelty Score | 0.206 ($p = 0.152$) |
| Number of Words in Title | 0.018 ($p < 0.001$) |
| Number of Words in Full Text (log) | 0.389 ($p < 0.001$) |
| Number of References | 0.002 ($p = 0.001$) |
| Number of Authors | 0.034 ($p < 0.001$) |
| Corr. Author No. Prior Publications | 0.000 ($p < 0.001$) |
| Corr. Author No. Prior Citations (log) | 0.026 ($p < 0.001$) |
| Corr. Author Gender (3 categories) | Y |
| Submission Year (2011–2022) | Y |
| Research Topics (14 categories) | Y |
| Intercept | −4.472 ($p < 0.001$) |
| Pseudo-$R^2$ | 0.032 |
| Observations | 44,528 |

TABLE S6. OLS regression model for predicting the **rating score** of an ICLR paper. We used each (paper, reviewer) pair as an observation. Coefficient estimates are reported with $p$-values shown in parentheses. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 1.5$).

| | *Dependent variable:* *Reviewer Rating Score* |
|---|---|
| Percentage of "We" Words | 26.160 ($p < 0.001$) |
| Percentage of Promotional Words | 23.685 ($p < 0.001$) |
| Flesch Reading Score | $-0.032$ ($p < 0.001$) |
| Concreteness Score | $-0.074$ ($p < 0.001$) |
| Novelty Score | 0.368 ($p = 0.002$) |
| Number of Words in Title | $-0.015$ ($p < 0.001$) |
| Number of Words in Full Text (log) | 1.350 ($p < 0.001$) |
| Number of References | 0.003 ($p < 0.001$) |
| Submission Year (2018–2024) | Y |
| Research Topics (7 categories) | Y |
| Intercept | $-7.065$ ($p < 0.001$) |
| $R^2$ | 0.128 |
| Observations | 70,365 |

TABLE S7. OLS regression model for predicting the **positivity score** (TextBlob) of a review comment for each ICLR paper. We used each (paper, reviewer) pair as an observation. Coefficient estimates are reported with $p$-values shown in parentheses. The positivity score is rescaled to the same range as the reviewer rating score (1–10) for better interpretability. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 1.5$).

| | *Dependent variable:* *Reviewer Positivity Score (TextBlob)* |
|---|---|
| Percentage of "We" Words | 3.372 ($p < 0.001$) |
| Percentage of Promotional Words | 9.075 ($p < 0.001$) |
| Flesch Reading Score | −0.010 ($p < 0.001$) |
| Concreteness Score | −0.002 ($p = 0.439$) |
| Novelty Score | −0.010 ($p = 0.794$) |
| Number of Words in Title | −0.003 ($p < 0.001$) |
| Number of Words in Full Text (log) | 0.120 ($p < 0.001$) |
| Number of References | 0.001 ($p < 0.001$) |
| Review Year (2018–2024) | Y |
| Research Topics (7 categories) | Y |
| Intercept | 5.279 ($p < 0.001$) |
| $R^2$ | 0.142 |
| Observations | 70,365 |

TABLE S8. OLS regression model for predicting the **positivity score** (SciBERT) of a review comment for each ICLR paper. We used each (paper, reviewer) pair as an observation. Coefficient estimates are reported with $p$-values shown in parentheses. The positivity score is rescaled to the same range as the reviewer rating score (1–10) for better interpretability. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues (all VIFs $< 1.5$).

| | *Dependent variable:* *Reviewer Positivity Score (SciBERT)* |
|---|---|
| Percentage of "We" Words | 0.151 ($p < 0.001$) |
| Percentage of Promotional Words | 0.418 ($p < 0.001$) |
| Flesch Reading Score | 0.000 ($p = 0.919$) |
| Concreteness Score | 0.000 ($p = 0.387$) |
| Novelty Score | 0.003 ($p = 0.375$) |
| Number of Words in Title | 0.000 ($p < 0.001$) |
| Number of Words in Full Text (log) | 0.002 ($p < 0.001$) |
| Number of References | 0.000 ($p < 0.001$) |
| Review Year (2018–2024) | Y |
| Research Topics (7 categories) | Y |
| Intercept | 0.019 ($p < 0.001$) |
| $R^2$ | 0.032 |
| Observations | 70,365 |

TABLE S9. OLS regression model for predicting the **engagement score** of a review comment for each ICLR paper. We used each (paper, reviewer) pair as an observation. Coefficient estimates are reported with $p$-values shown in parentheses. The engagement score is rescaled to the same range as the reviewer's rating score (1–10) for better interpretation. Variance Inflation Factor (VIF) check indicates no concerning multicollinearity issues among predictor variables (all VIFs $< 1.5$).

| | *Dependent variable:* *Reviewer Engagement Score* |
|---|---|
| Percentage of "We" Words | 2.640 ($p < 0.001$) |
| Percentage of Promotional Words | 6.357 ($p < 0.001$) |
| Flesch Reading Score | 0.011 ($p = 0.002$) |
| Concreteness Score | −0.007 ($p = 0.037$) |
| Novelty Score | 0.176 ($p = 0.004$) |
| Number of Words in Title | −0.001 ($p = 0.375$) |
| Number of Words in Full Text (log) | 0.095 ($p < 0.001$) |
| Number of References | −0.001 ($p < 0.001$) |
| Review Year (2018–2024) | Y |
| Research Topics (7 categories) | Y |
| Intercept | 1.247 ($p < 0.001$) |
| $R^2$ | 0.028 |
| Observations | 70,365 |

TABLE S10. The causal mediation analysis based on our LLM-assisted experimental data using all ICLR abstracts. Two versions (original and without-"We" replica) of each abstract were independently evaluated by LLM agents on Decision Score (our dependent variable) and four potential mediators, including Credibility, Significance, Robustness, and Teamwork. Coefficients of the total effect model and the direct effect model are reported below with $p$-values shown in parentheses. Teamwork has the largest impact on Decision Score, followed by Credibility, Significance, and Robustness. The mediator models are reported in Table S11 for space consideration. All models control for four linguistic features at the abstract level.

| | *Dependent variable: Decision Score* | |
|---|---|---|
| | **Total Effect Model** | **Direct Effect Model** |
| Percentage of "We" Words | 43.273 ($p < 0.001$) | 1.998 ($p < 0.001$) |
| Percentage of Promotional Words | 7.158 ($p < 0.001$) | −0.123 ($p = 0.345$) |
| Flesch Reading Score | −0.009 ($p < 0.001$) | 0.001 ($p < 0.001$) |
| Concreteness Score | −1.034 ($p < 0.001$) | −0.023 ($p = 0.034$) |
| Number of Words in Abstract (log) | 1.422 ($p < 0.001$) | −0.002 ($p = 0.749$) |
| Credibility Score | | 0.327 ($p < 0.001$) |
| Significance Score | | 0.292 ($p < 0.001$) |
| Robustness Score | | 0.112 ($p < 0.001$) |
| Teamwork Score | | 0.401 ($p < 0.001$) |
| Intercept | 0.851 ($p < 0.001$) | −0.349 ($p < 0.001$) |
| $R^2$ | 0.278 | 0.963 |
| Observations | 38,508 | 38,508 |

TABLE S11. The causal mediation analysis shows that the density of "We" words strongly predicts each of the four mediators. Coefficients of our mediator models are reported below with $p$-values shown in parentheses. The effect of "We" is most pronounced for Credibility, followed by Robustness, Teamwork, and Significance.

| | *Dependent variable:* | | | |
|---|---|---|---|---|
| | **Credibility** | **Robustness** | **Teamwork** | **Significance** |
| Percentage of "We" Words | 40.729 ($p < 0.001$) | 40.200 ($p < 0.001$) | 34.233 ($p < 0.001$) | 33.402 ($p < 0.001$) |
| **Controls** | | | | |
| Percentage of Promotional Words | 4.658 ($p < 0.001$) | 6.344 ($p < 0.001$) | 7.256 ($p < 0.001$) | 7.342 ($p < 0.001$) |
| Flesch Reading Score | −0.008 ($p < 0.001$) | −0.008 ($p < 0.001$) | −0.009 ($p < 0.001$) | −0.011 ($p < 0.001$) |
| Concreteness Score | −1.275 ($p < 0.001$) | −1.146 ($p < 0.001$) | −0.631 ($p < 0.001$) | −0.733 ($p < 0.001$) |
| Number of Words in Abstract (log) | 1.299 ($p < 0.001$) | 1.227 ($p < 0.001$) | 1.226 ($p < 0.001$) | 1.269 ($p < 0.001$) |
| $R^2$ | 0.273 | 0.316 | 0.281 | 0.200 |
| Observations | 38,508 | 38,508 | 38,508 | 38,508 |

## II. SUPPLEMENTARY TEXT

### A. Dictionary of Promotional Words

The table below lists the full lexicon of 139 scientific promotional words grouped into 8 thematic categories. The lexicon was developed by a prior study [1] and validated by our prior work [2, 3].

- **Importance:**
  *compelling, critical, crucial, essential, foundational, fundamental, imperative, important, indispensable, invaluable, key, major, paramount, pivotal, significant, strategic, timely, ultimate, urgent, vital*

- **Novelty:**
  *creative, emerging, first, groundbreaking, innovative, latest, novel, revolutionary, unique, unparalleled, unprecedented*

- **Rigor:**
  *accurate, advanced, careful, cohesive, detailed, nuanced, powerful, quality, reproducible, rigorous, robust, scientific, sophisticated, strong, systematic*

- **Scale:**
  *ample, biggest, broad, comprehensive, considerable, deeper, diverse, enormous, expansive, extensive, fastest, greatest, huge, immediate, immense, interdisciplinary, international, interprofessional, largest, massive, multidisciplinary, myriad, overwhelming, substantial, top, transdisciplinary, tremendous, vast*

- **Utility:**
  *accessible, actionable, deployable, durable, easy, effective, efficacious, efficient, generalizable, ideal, impactful, intuitive, meaningful, productive, ready, relevant, rich, safer, scalable, seamless, sustainable, synergistic, tailored, tangible, transformative, user-friendly*

- **Quality:**
  *ambitious, collegial, dedicated, exceptional, experienced, intellectual, longstanding, motivated, premier, prestigious, promising, qualified, renowned, senior, skilled, stellar, successful, talented, vibrant*

- **Attitude:**
  *attractive, confident, exciting, incredible, interesting, intriguing, notable, outstanding, remarkable, surprising*

- **Problem:**
  *alarming, daunting, desperate, devastating, dire, dismal, elusive, stark, unanswered, unmet*

### B. Dictionary of Engagement Words

The table below lists the lexicon of all 116 scientific engagement words identified in [4].

*absorbing, address, agreeable, allure, alluring, appealing, appreciate, arresting, attach, attract, attracting, attraction, attractive, be a party to, be absorbed in, be associated with, become involved in, captivate, captivating, captivation, charismatic, charm, charming, collaborate, concentrated upon, concern, concerned with, congenial, cooperate, cute, delightful, embark on, enamour, enchant, enchanting, enchantment, endearing, engage, engaged, engaging, engross, engrossed, engrossing, engulf, enjoy, enjoyable, entertaining, enticing, entrancing, enwrap, espousal, exciting, fascinate, fascinating, fill up, get involved in, glamorous, go in for, gorgeous, gratifying, heavenly, immerse, improve, interest, interesting, intrigue, intriguing, inviting, involve, involvement, likable, likeable, lovable, lovely, magnetic, mesmeric, mesmerizing, now, obsess, obsessed, obsession, occupying, partake, partake in, partake of, participate in, participation, play a part in, play a role in, pleasant, pleasing, pleasurable, point, preoccupy, prepossessing, pretty, previous, provocative, raise, recommend, response, revise, satisfying, seductive, set about, share, share in, siren, smashing, still, sweet, take on, take part in, take upon oneself, tempting, version*

## C. Rating Task Guideline for Human Raters

### 1. Task Overview

This task aims to collect human ratings of academic papers. As an experienced researcher specializing in AI, NLP, and machine learning, you are asked to review the abstract of 100 papers submitted to a top AI conference. Please carefully evaluate each abstract and provide a rating score on five dimensions using your expert knowledge and judgment. We estimate that each paper takes about 5 minutes for your review.

### 2. Evaluation Criteria Commonly Used in Peer Review

| **Dimension** | **Explanation** |
|---|---|
| **Credibility** | How credible is this paper?<br>Credibility refers to the trustworthiness, validity, and reliability of the research, including its methods, findings, and interpretations. A higher score indicates stronger scientific rigor and sound methodology. |
| **Significance** | How significant is the research problem addressed?<br>Significance captures the importance, relevance, and potential impact of the work—whether it contributes meaningfully to its field or broader scientific understanding. A higher score reflects a more important and timely contribution. |
| **Robustness** | How robust do the results and findings appear?<br>Robustness assesses the reproducibility, generalizability, and methodological strength of the study under different settings or datasets. A higher score indicates greater confidence in the reliability of the results. |
| **Teamwork** | To what extent does the paper reflect collaboration and team effort?<br>Teamwork is inferred from authorship patterns, project scope, and writing style that suggest collective input. A higher score suggests the work is likely a product of effective collaboration. |
| **Decision Score** | How likely are you to recommend this paper for acceptance?<br>This is an overall recommendation score, based on your synthesis of the above four aspects and any other information inferred from the abstract. |

**Instruction:** Please read each abstract carefully and base your evaluation solely on the content of the abstract before assigning scores. You can provide a numeric score (1–10) for each dimension.

**Your ratings:**

| **Dimension** | **Score (1–10)** |
|---|---|
| Credibility | |
| Significance | |
| Robustness | |
| Teamwork | |
| Decision Score | |

[1] Neil Millar, Bojan Batalo, and Brian Budgell. Trends in the use of promotional language (hype) in abstracts of successful national institutes of health grant applications, 1985-2020. *JAMA Network Open*, 5(8):e2228676–e2228676, 2022.

[2] Hao Peng, Huilian Sophie Qiu, Henrik Barslund Fosse, and Brian Uzzi. Promotional language and the adoption of innovative ideas in science. *Proceedings of the National Academy of Sciences*, 121(25):e2320066121, 2024.

[3] Huilian Sophie Qiu, Hao Peng, Henrik Barslund Fosse, , Teresa Woodruff, and Brian Uzzi. Use of promotional language in grant applications and grant success. *JAMA Network Open*, 7(12):e2448696, 2024.

[4] Zhiwen Sun, Changcheng Cao, Sheng Liu, et al. Behavioral consequences of second-person pronouns in written communications between authors and reviewers of scientific papers. *Nature Communications*, 15:152, 2024. doi: 10.1038/s41467-023-44515-1. URL `https://doi.org/10.1038/s41467-023-44515-1`.